PISA scoring and PISA levels of proficiency (from T.Benton)

1 message

Aryan Thanki <aryanthanki313@gmail.com> 8 February 2024 at 17:47
To: tompiccolo50@gmail.com

---------- Forwarded message ---------
From: **Tom Benton** <tom.benton@cambridge.org>
Date: Thu, 8 Feb 2024 at 09:59
Subject: RE: How are PISA scores calculated
To: Aryan Thanki <aryanthanki313@gmail.com>

Hello Aryan

These are complicated questions. I will do my best to provide answers.

How are PISA score calculated

This is very complicated. The full procedure is documented here (https://www.oecd.org/pisa/data/pisa2022technicalreport/PISA-2022-Technical-Report-Ch-11-Scaling-PISA-Data.pdf).

Broadly, and briefly, the whole procedure is designed to measure the performance of nations rather than individual students. The distribution of scores for nations will (pretty much) be entirely defined by how well the students in that nation performed on the relevant test questions. For example, a nations scores in mathematics will be determined by how well their students performed on the mathematics items.

The scores for individual students are just guesses of how good each student is at each subject based on everything that is known about them. This will include how they performed on the relevant test (if they took it) but also relate to how well they did on other tests, their questionnaire responses and which school they attend. Like I said. It's complicated.

The scores for individual students are actually not intended for making inferences about individuals at all. They are just constructed in a way that allows further analysis of how ability in each subject relates to the other things that were measured within the study (e.g., in the questionnaires). They are also not designed to be used in analysis of data collected outside of the PISA study itself (e.g., GCSE scores).

What do PISA ratings mean

The main PISA report provide some descriptions of what students can do at different performance levels. These can be found on page 105 (Table I.6.1) of this report (https://www.oecd-ilibrary.org/education/pisa-2018-results-volume-i_5f07c754-en).

How can PISA measures academic ability along with GCSE examinations?

Being good at one set of examinations tends to be related to being good at another. So different tests will all tend to capture this.

However, PISA and GCSE have very different purposes.

PISA is designed and scores to capture the performance of nations.

GCSEs are designed and scored to capture the performance of individual students.

I hope that helps. If you want more detail there is an enormous amount to read on the web about how PISA works and what it measures. I will not be entering into a long email conversation about all this.

Best wishes

Tom

On Thu, 8 Feb 2024 at 10:38, Tom Benton <tom.benton@cambridge.org> wrote:

> You can see some examples of questions that students are likely to get correct at different levels of ability starting on page 125 of this report
>
> **"Climbing Mount Fuji" question 2 is an example of a question that someone with a PISA ability of 642 would be able to answer.**
>
> I need to do other work now and will not be replying to any further emails.
>
> Tom